

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	7
	3.3 SAR Based Analysis at Abutment 1 LHS	8
	3.4 SAR Based Analysis at Abutment 1 RHS	9
	3.5 SAR Based Analysis at Pier 1 LHS	10
	3.6 SAR Based Analysis at Pier 1 RHS	11
	3.7 SAR Based Analysis at Pier 2 LHS	12
	3.8 SAR Based Analysis at Pier 2 RHS	13
	3.9 SAR Based Analysis at Abutment 2 LHS	14
	3.10 SAR Based Analysis at Abutment 2 RHS	15
3.	Analysis	16
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

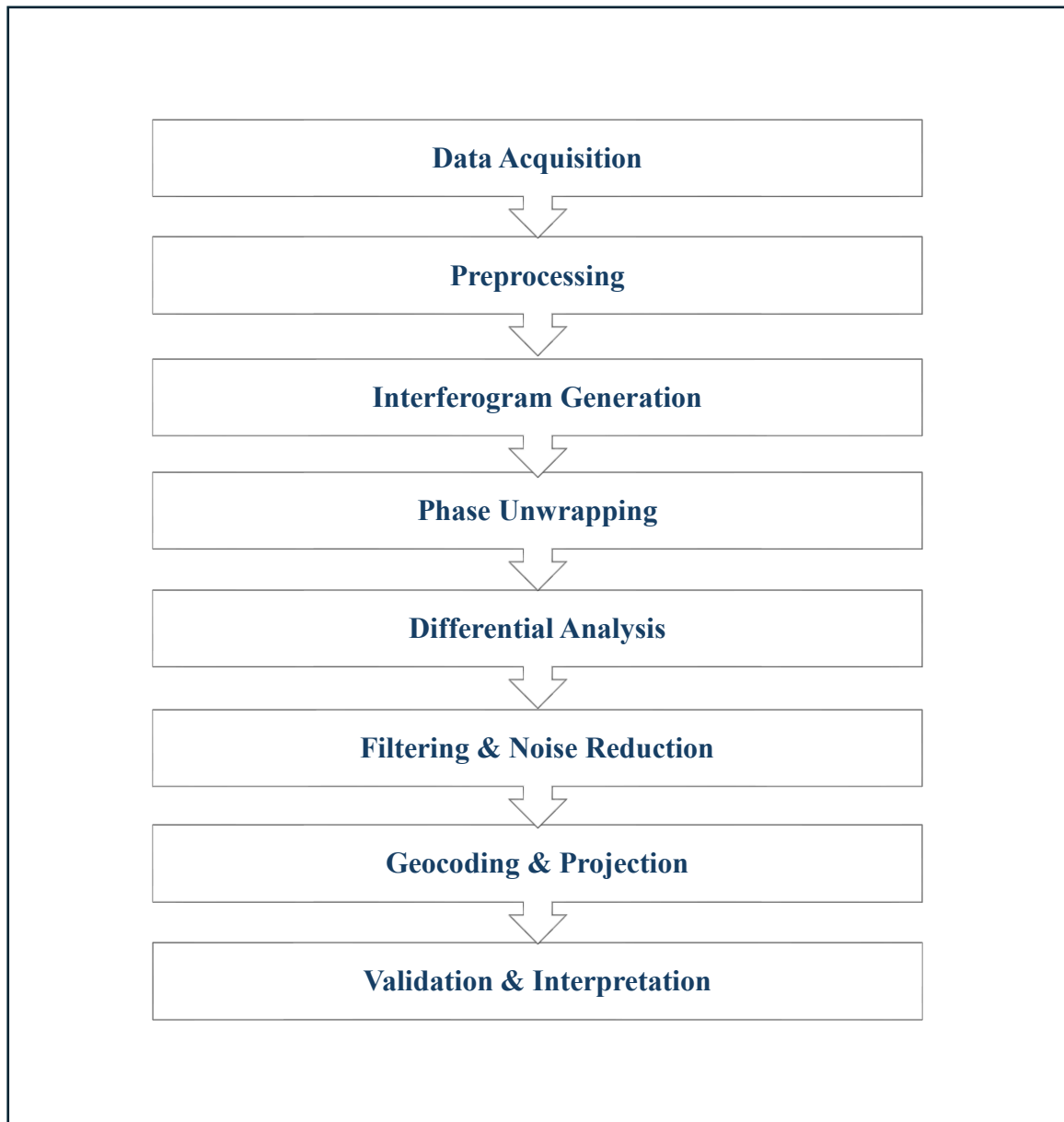
Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 946+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MJB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

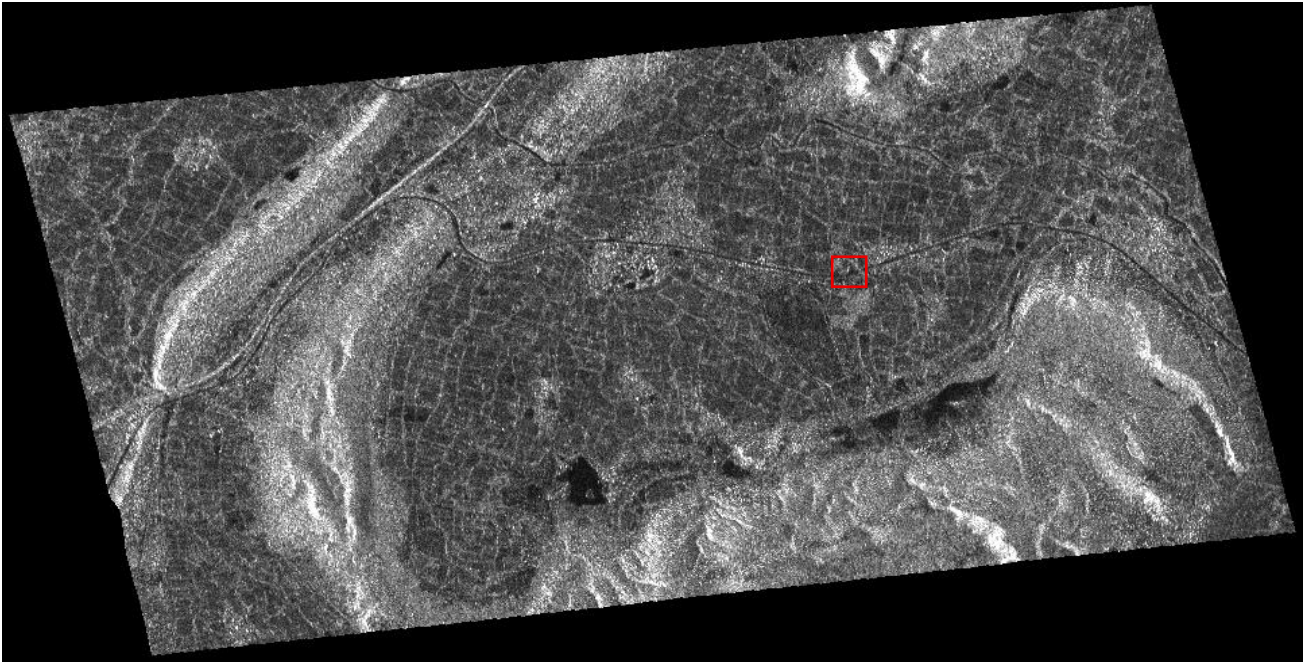
Flight Direction - DESCENDING

Polarization – VV+VH

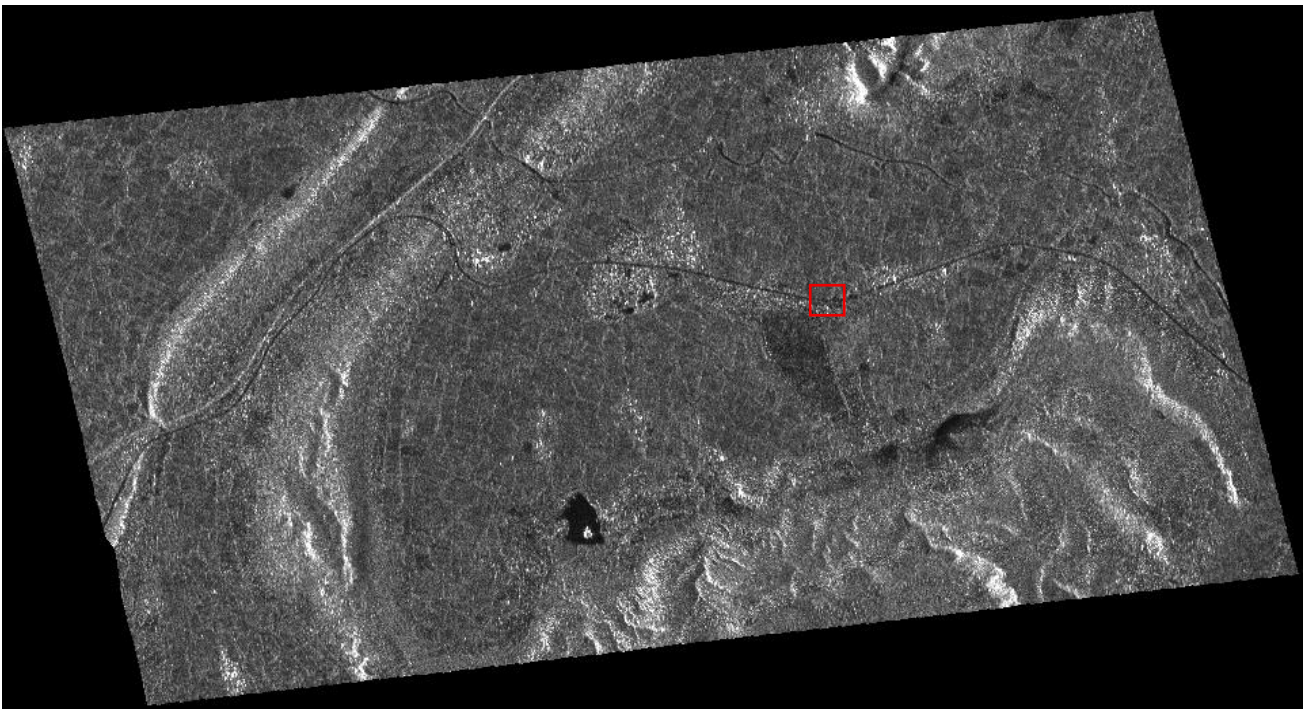
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

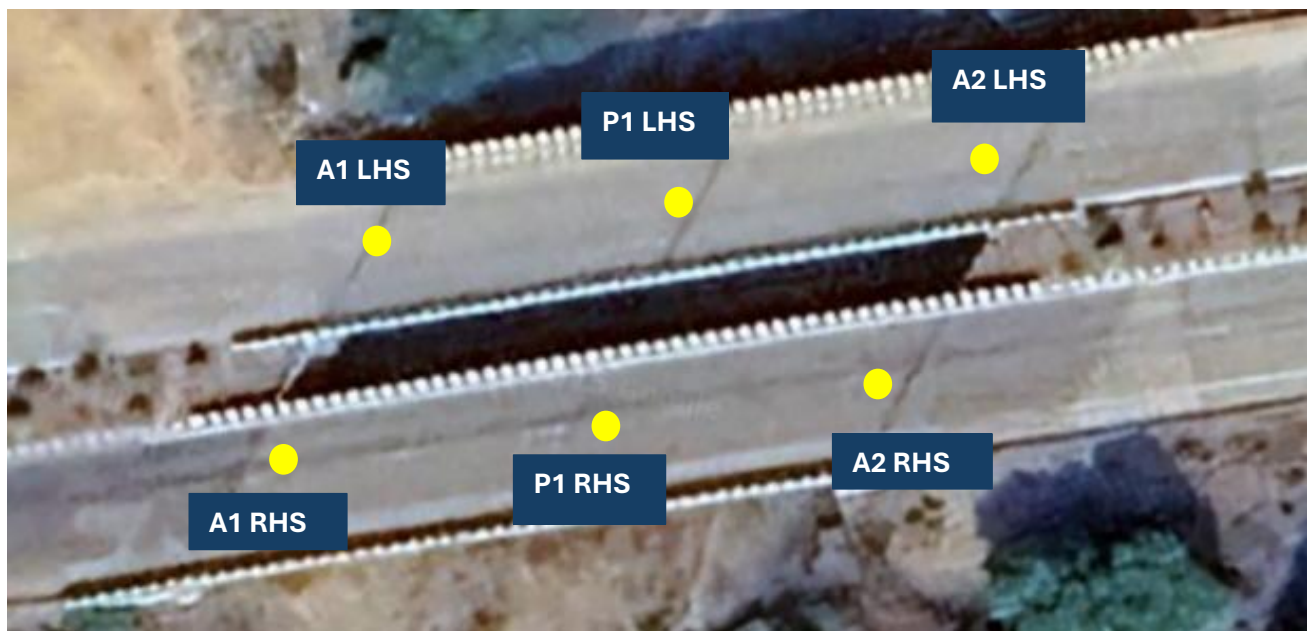


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-946+200

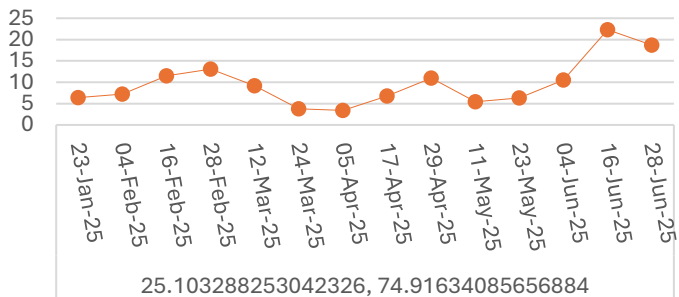
Length – 54.46M



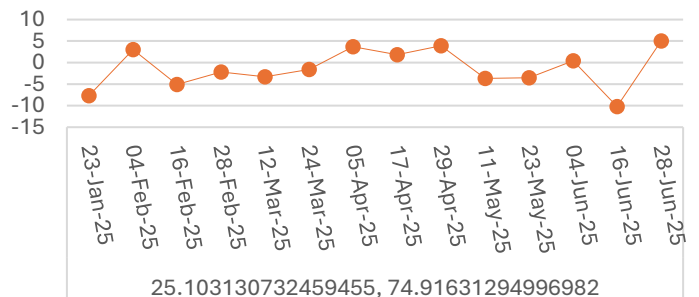
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

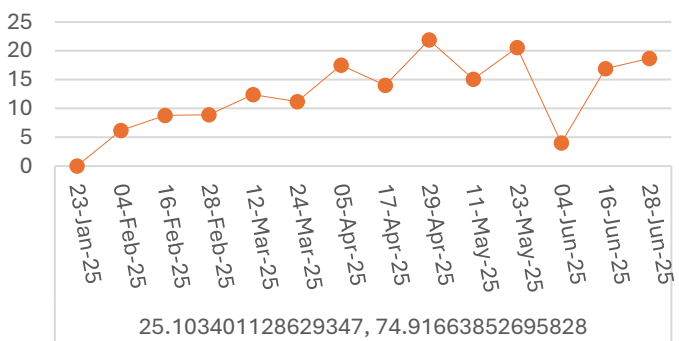
Displacement @ A1 LHS VV



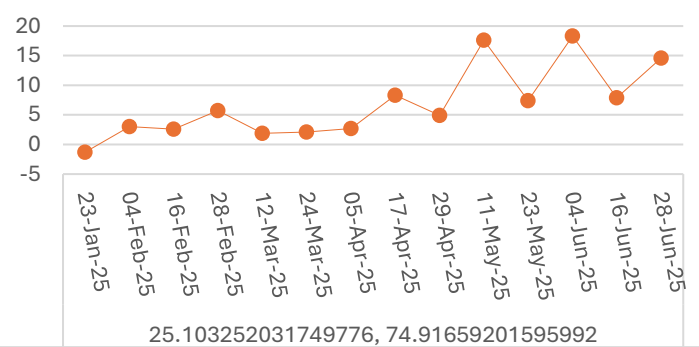
Displacement @ A1 RHS VV



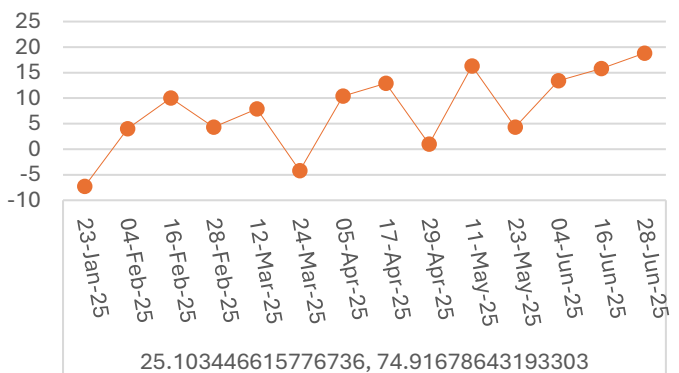
Displacement @ P1 LHS VV



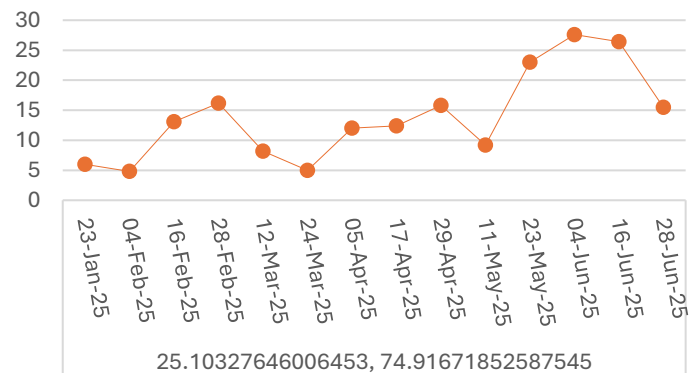
Displacement @ P1 RHS VV



Displacement @ A2 LHS VV

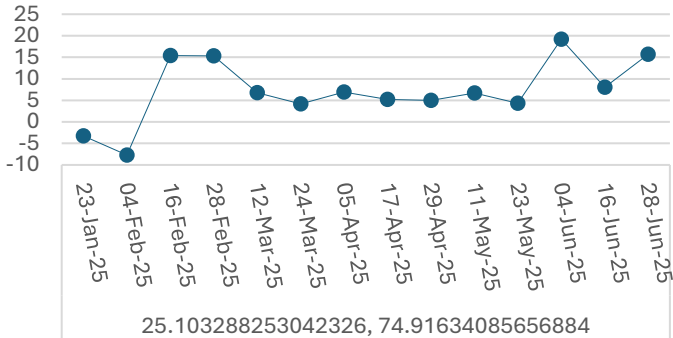


Displacement @ A2 RHS

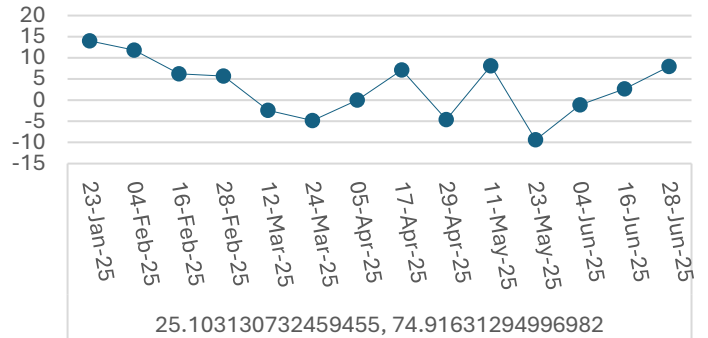


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

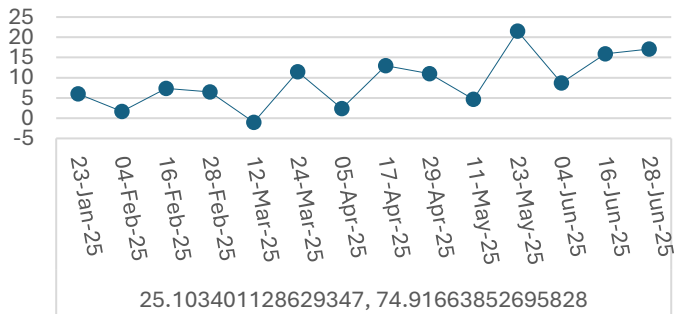
Displacement @ A1 LHS VH



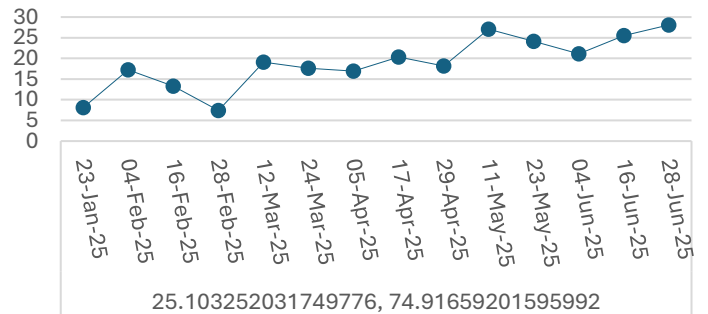
Displacement @ A1 RHS VH



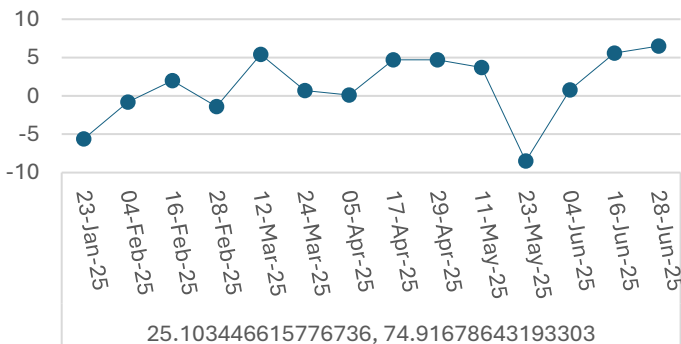
Displacement @ P1 LHS VH



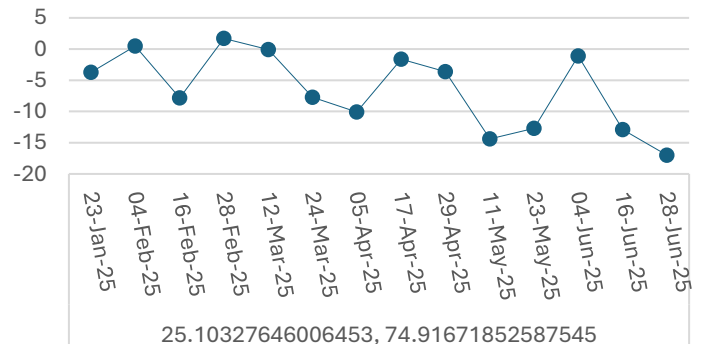
Displacement @ P1 RHS VH



Displacement @ A2 LHS VH



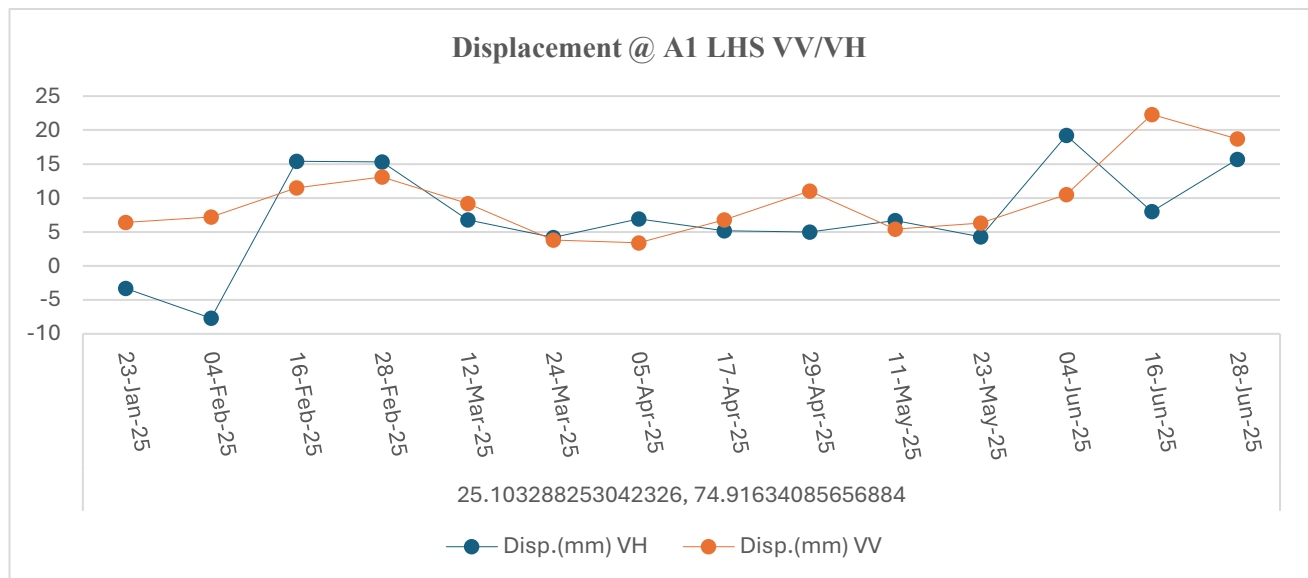
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.103288253042326, 74.91634085656884)



Key Observations:

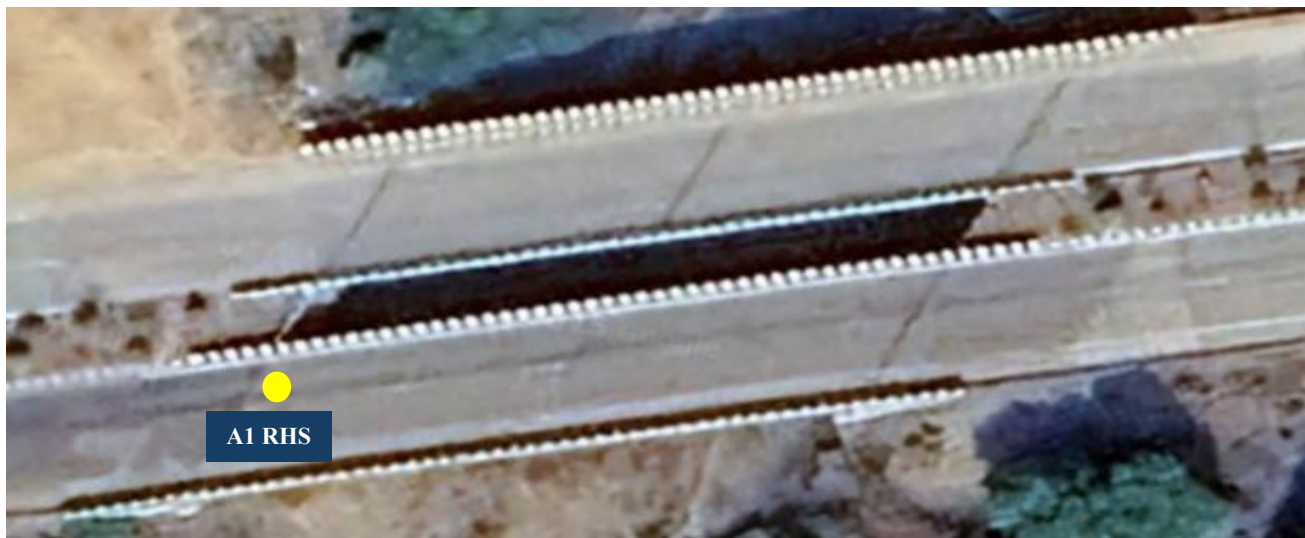
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a generally upward trend over the monitoring period, varying from approximately -8 mm to +22 mm. The movements indicate a progressive vertical rise with intermittent drops, reflecting fluctuating but overall increasing upward displacement.
- **Horizontal displacement (VH)** fluctuates within the range of -5 mm to +17 mm, showing moderate lateral variability. The pattern indicates that horizontal movement remains active throughout the period, with noticeable peaks during late April and mid-June.

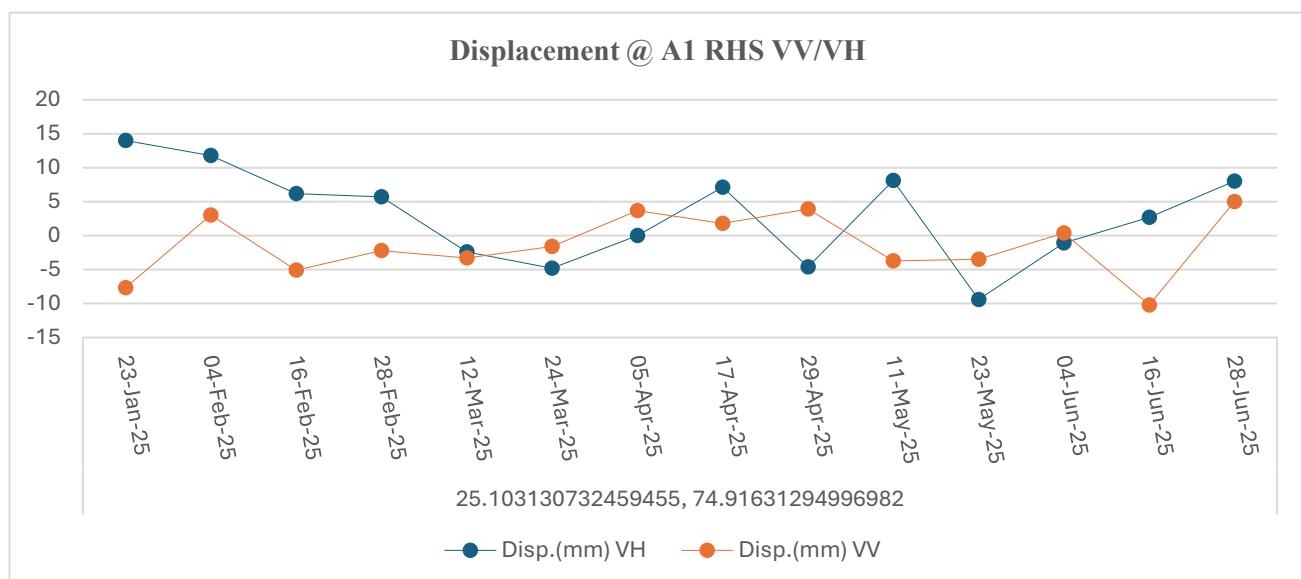
Critical Events & Anomalies:

- **February 2025:** Both VH and VV experience a sharp downward dip, with VV reaching around -8 mm and VH near -5 mm marking the first significant combined downward excursion.
- **March 2025:** VV shows a sustained rise, moving from around 0 mm to +13 mm. VH also increases
- **June 2025:** A temporary decline in both VH and VV occurs, followed by another upward shift at the end of June.

- 3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.103130732459455, 74.91631294996982)



Key Observations:

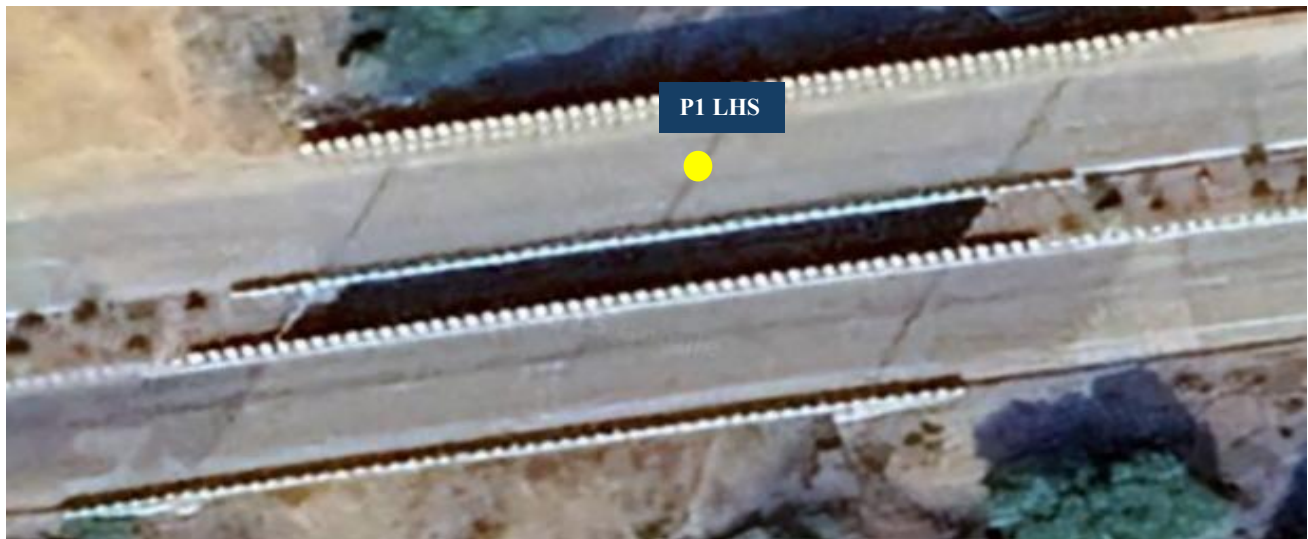
Displacement Magnitude Patterns:

- Horizontal displacement (VH)** varies from about **-10 mm to +15 mm**, reflecting **frequent lateral fluctuations** with no prolonged directional trend but clear episodic deviations.
- Vertical displacement (VV)** ranges between approximately **-12 mm and +6 mm**, showing alternating upward and downward movements, indicating a moderate but non-uniform vertical response throughout the monitoring period.

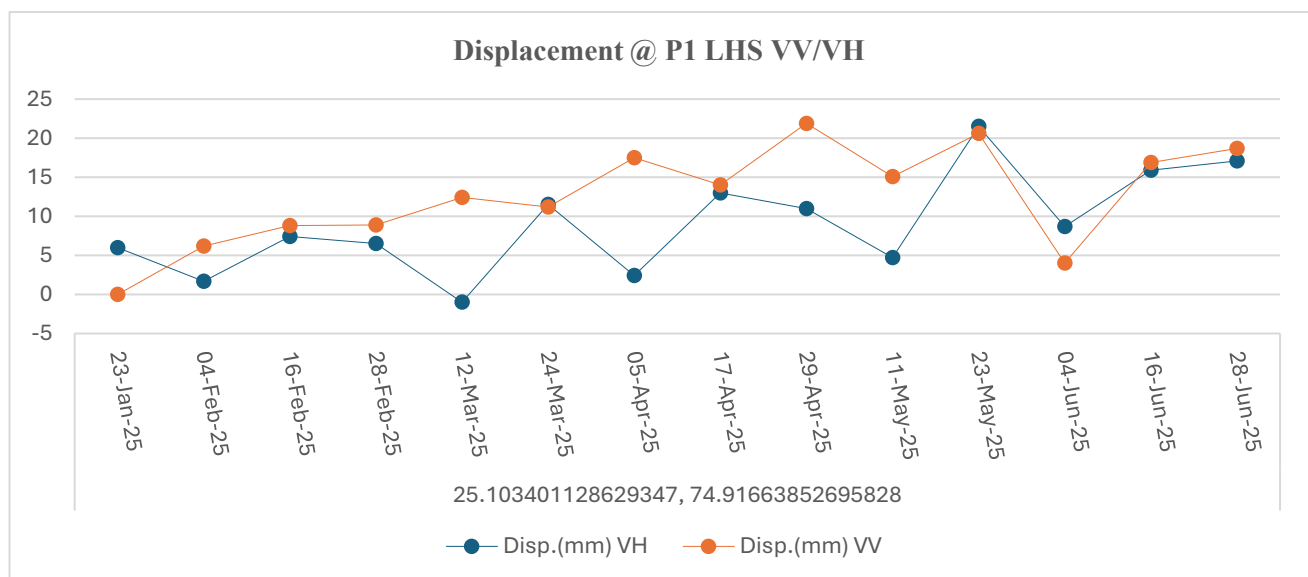
Critical Events & Anomalies:

- Jan-Feb 2025:** VH starts high before gradually reducing to around +7 mm, while VV shifts from -8 mm to +4 mm, marking an early-phase vertical correction.
- March 2025:** Both VH and VV move near the neutral line indicating a short period of relative stability
- June 2025:** Both VV and VH climb again toward +5 to +10 mm, indicating renewed positive displacement on both axes.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.103401128629347, 74.91663852695828)



Key Observations:

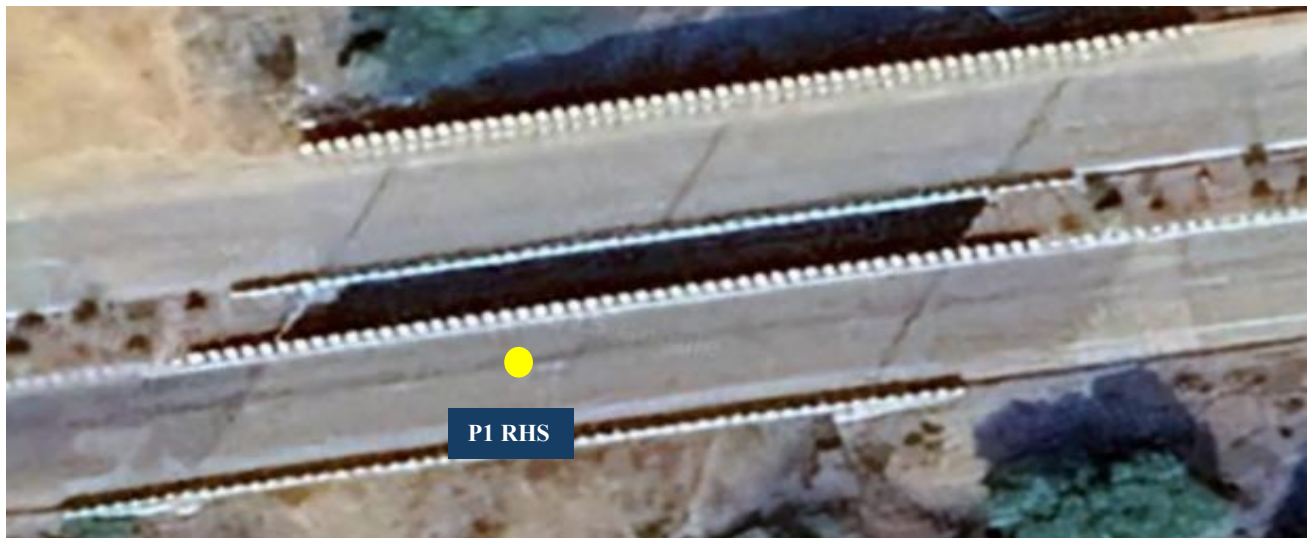
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV shows a clear upward trend, starting in January and increasing steadily by June 2025. This reflects progressive upward movement/lifting of the monitored point over time.
- **Horizontal Displacement (VH)** VH fluctuates between 0 mm and +20 mm, showing moderate but increasing lateral movement.

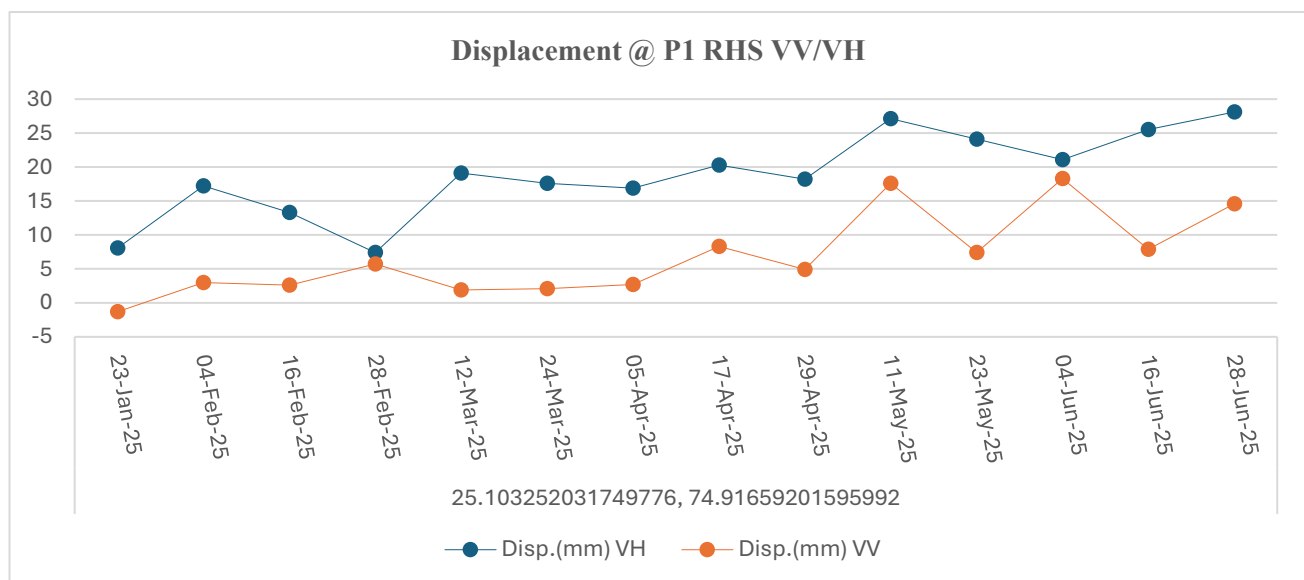
Critical Events & Anomalies:

- **January–February 2025:** Early VH movement remains low (0–7 mm), while VV rises gradually from 0 mm to about +7 mm, indicating the beginning of upward displacement.
- **May 2025:** V peaks near +22 mm, the maximum vertical displacement recorded, while VH simultaneously increases to +15–20 mm, indicating a combined vertical and lateral movement spike.
- **June 2025:** Displacements stabilize at high values, with VV sustaining +17–20 mm and VH holding +15–18 mm, suggesting continued deformation without reversal.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.103252031749776, 74.91659201595992)



Key Observations:

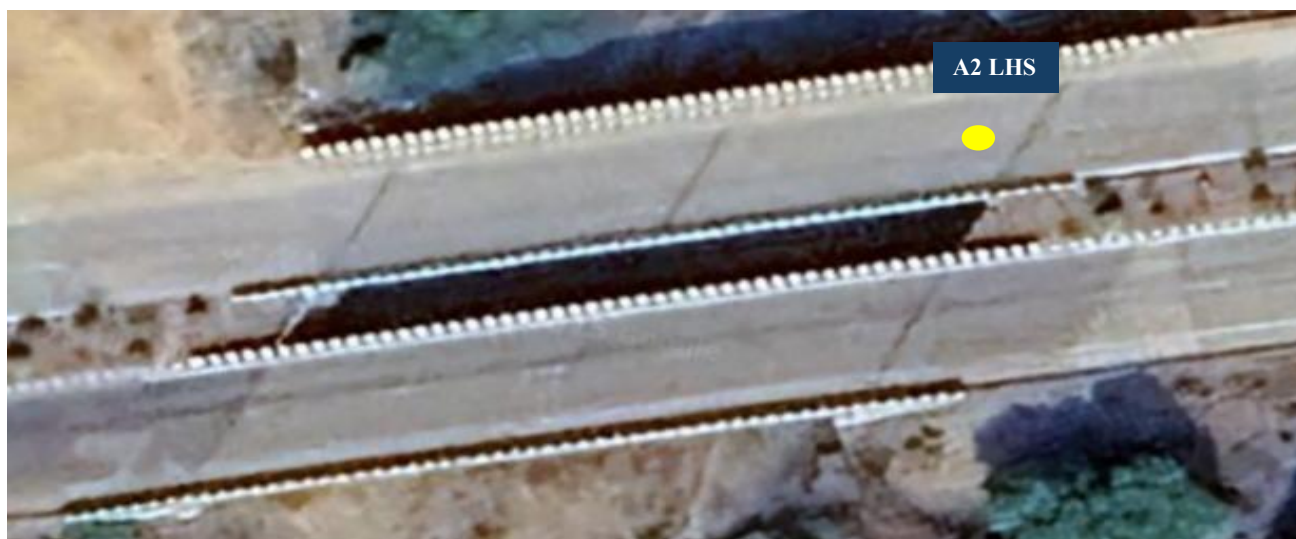
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV shows moderate upward movement overall, ranging between –5 mm to +18 mm
- **Horizontal Displacement (VH)** VH demonstrates significant positive displacement, consistently ranging from +8 mm to +28 mm, indicating sustained lateral movement in one dominant direction.

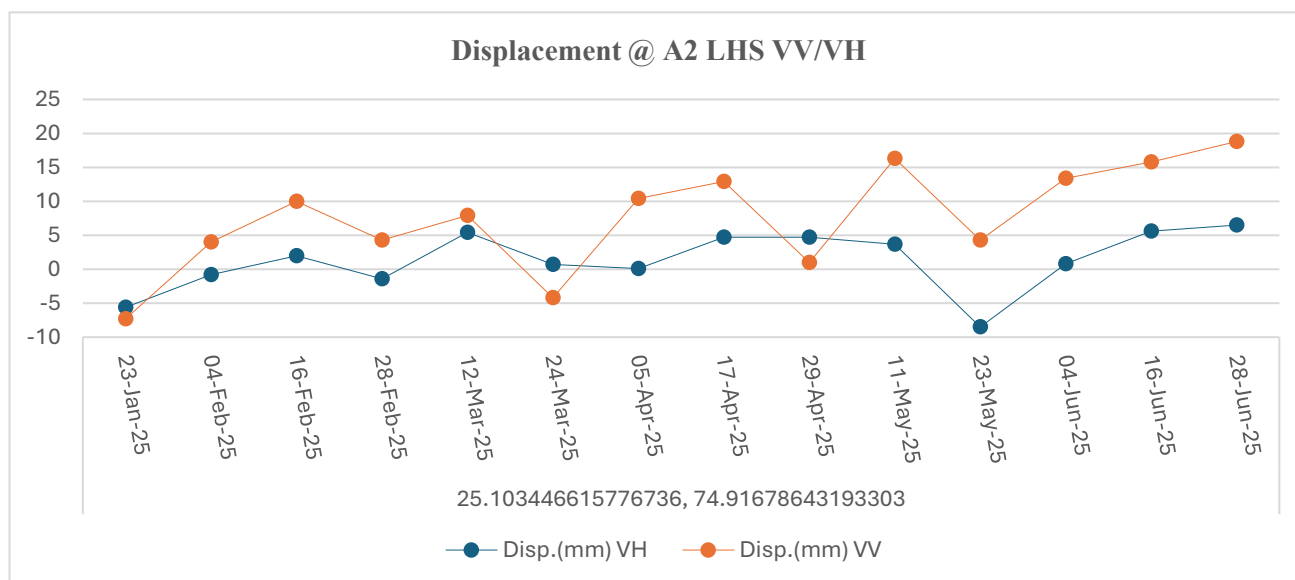
Critical Events & Anomalies:

- **Feb 2025:** Horizontal displacement rises sharply from about +8 mm to +16 mm, marking the first significant lateral shift.
- **March 2025:** VV drops to approximately –1 mm followed by a rebound to +8 mm, indicating a short-term vertical instability.
- **June 2025:** Both VV (up to +15 mm) and VH (around +28 mm) remain elevated, indicating a continuing deformation trend rather than stabilization.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.103446615776736, 74.91678643193303)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a clear upward trend over the monitoring period, increasing from approximately – 5 mm in January 2025 to around +20 mm by June 2025. This indicates a progressive upward movement or recovery/lifting at the A2 LHS location.
- **Horizontal Displacement (VH)** VH fluctuates moderately between –10 mm and +10 mm, with a brief downward spike in late May. Overall, VH shows small but consistent lateral variations without long-term directional drift.

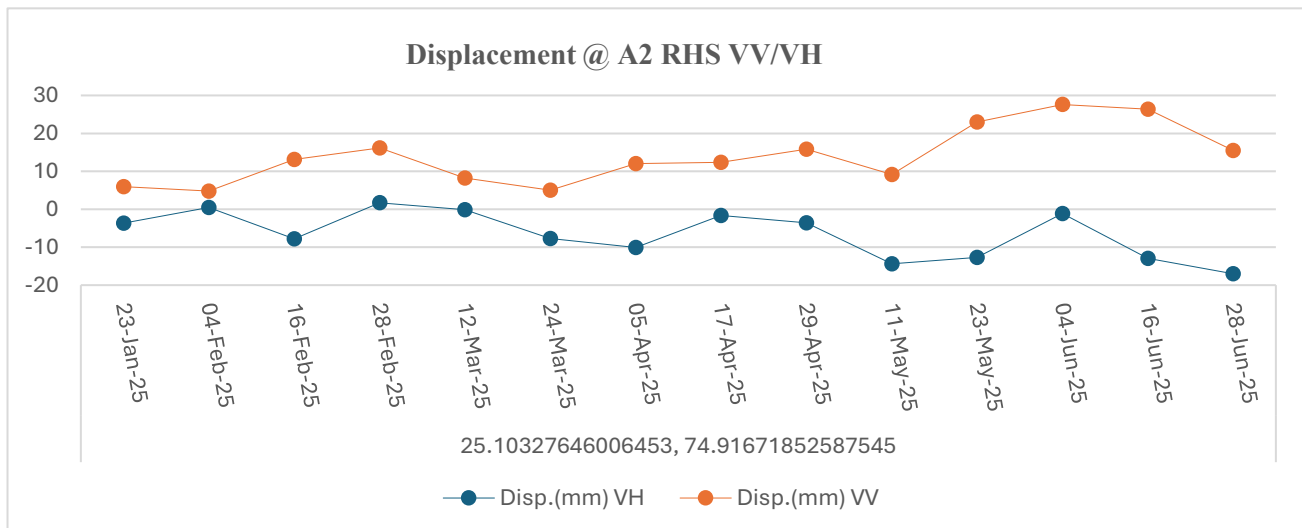
Critical Events & Anomalies:

- **Jan-Feb 2025:** Both VH and VV remain within ± 5 –10 mm, showing only minor early-stage variations.
- **June 2025:** Both components rise steadily, with VV reaching its highest values (+20 mm) and VH stabilizing between +5 and +8 mm, suggesting increasing vertical deformation with controlled lateral movement.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.10327646006453, 74.91671852587545)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a clear upward-trending behaviour, fluctuating between +5 mm and +28 mm, indicating progressive vertical uplift on the RHS with several pronounced peaks
- **Horizontal Displacement (VH)** exhibits predominantly negative (southward) movement ranging from –18 mm to +5 mm, highlighting consistent lateral drift with repeated downward spikes.

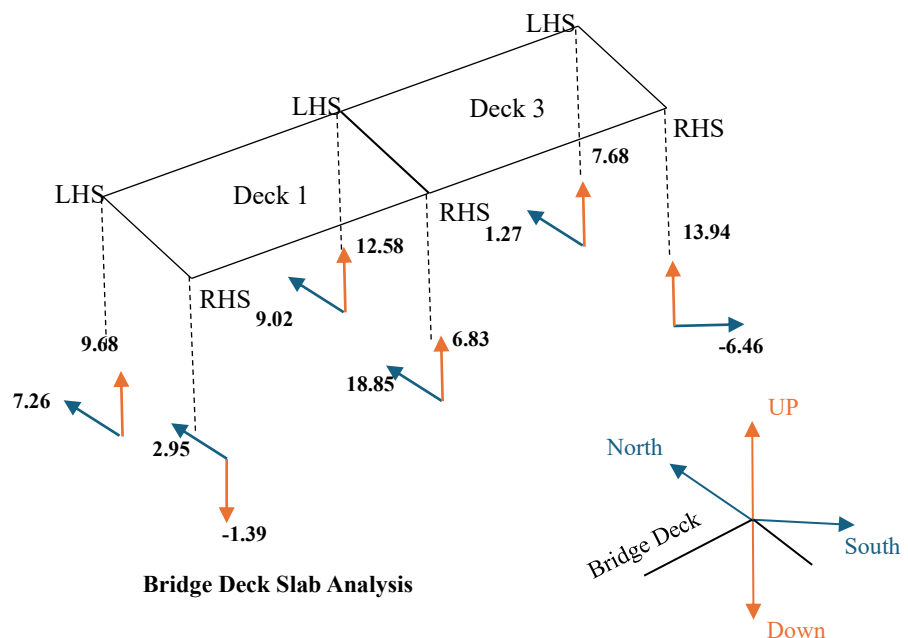
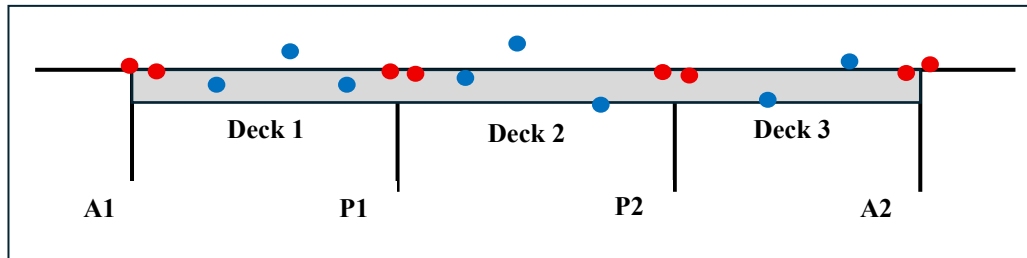
Critical Events & Anomalies:

- **February 2025:** VV rises to around +15 mm, representing the first major vertical uplift phase, while VH dips to –10 mm, marking a noticeable lateral shift.
- **May 2025:** VV reaches a significant peak of +27 mm, its highest recorded value within the monitoring period.
- **June 2025:** VH drops sharply to –15 mm, representing the largest lateral deviation observed. VV again approaches +20 mm, indicating sustained upward displacement rather than recovery.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The vertical displacement values across Decks 1 and 3 exhibit a mix of upward (positive) and downward (negative) responses, indicating typical deck-slab flexibility under service loads. Overall, the magnitudes remain moderate, with no excessive sagging or uplift that would indicate abnormal distress.

Horizontal Displacement Behaviour:

- Horizontal displacement vectors show small to moderate shifts predominantly oriented toward the north and north-east, consistent with normal deck-shift trends in multi-span bridges. The directional vectors reveal a smooth displacement progression from Deck 1 to Deck 3, indicating a coordinated movement pattern rather than isolated deformation.

CONCLUSION:

- The combined vertical and horizontal displacements across all monitored deck points suggest that the bridge deck system is behaving normally, with no indications of structural instability or torsional distortion. The observed uplift, settlement, and lateral movement patterns align well with seasonal temperature variations, material expansion-contraction cycles, and normal traffic-induced dynamic effects. No abnormal concentration of deformation is present, and the displacement fields across the decks remain uniform and structurally acceptable.